



BUSINESS BLUEPRINT AOP AUTOMATION FOR HMD MEG PLANT

Honeywell

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Business Blueprint

AOP Automation for HMD MEG Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

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Revision History

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Table of Contents

List of Figures

Figure 1 Digitized AOP Flow Diagram	8
Figure 2 Digitized AOP screen for Production & Norms Basis	10
Figure 3 Digitized AOP Steps to derive EOE and EO month wise MC values	11
Figure 4 Digitized AOP screen for MEG Yield	12
Figure 5 Digitized AOP screen for Production Target.....	12
Figure 6 Digitized AOP screen for Business Demand	13
Figure 7 Digitized AOP screen for Shutdown/Turnaround Activities	13
Figure 8 Digitized AOP screen for Slowdown Activities.....	14
Figure 9 AOP Maintenance Details Table.....	16
Figure 10 Digitized AOP screen for Month wise Production plan	17
Figure 11 Digitized AOP Steps to derive month wise Historical Best Norms	18
Figure 12 Digitized AOP screen for Catalyst Selectivity month wise	19
Figure 13 Digitized AOP screen for Steady State Consumption	20
Figure 14 Digitized AOP screen for Shutdown Consumption	21
Figure 15 Digitized AOP screen for Slowdown Consumption.....	22
Figure 16 Digitized AOP screen for Budget Norms or Overall AOP Consumption.....	23
Figure 17 Digitized AOP screen for Annual AOP Cost	24
Figure 18 Digitized AOP screen for Production Plant Summary	25
Figure 19 Digitized AOP screen for Month Wise Production Plan.....	25
Figure 20 Digitized AOP screen for Month Wise Raw Data	26
Figure 21 Digitized AOP screen for Turnaround Report.....	26
Figure 22 Digitized AOP screen for Annual Production Plan.....	27
Figure 23 Digitized AOP screen for Plant Contribution.....	27

List of Tables

Table 1 Reference Documents	7
Table 2 AOP Functional Scope.....	8
Table 3 Maintenance Details	15
Table 4 Raw Material Norms calculations.....	19

1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for HMD MEG plant for HMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various MEG plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for MEG technology.

1.2 Reference Documents

Document Name	Revision	Title	Document
HMD MEG DCT	-	Digital AOP HMD MEG DCT	 DCT HMD MEG CHANGES REV 2.xlsx
UAT Observation list	-	Digital AOP MEG UAT Observation list	 Digital AOP MEG UAT Observations.xlsx
EOE effective Operating Hours calculation	-	RIL AOP - MEG Maintenance Screen - Slowdown Logic_R&T Validated	 RIL AOP - MEG Maintenance Screen
Slowdown Norms Methodology	-	Digital AOP - EOEG - Normal_slowdown_shutdown_Sample Norms Calculation	 Digital AOP - EOEG - Normal_slowdown_sh

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Overall AOP Consumption

Steps to generate the Production and Overall AOP Consumption are captured in the below flow diagram.

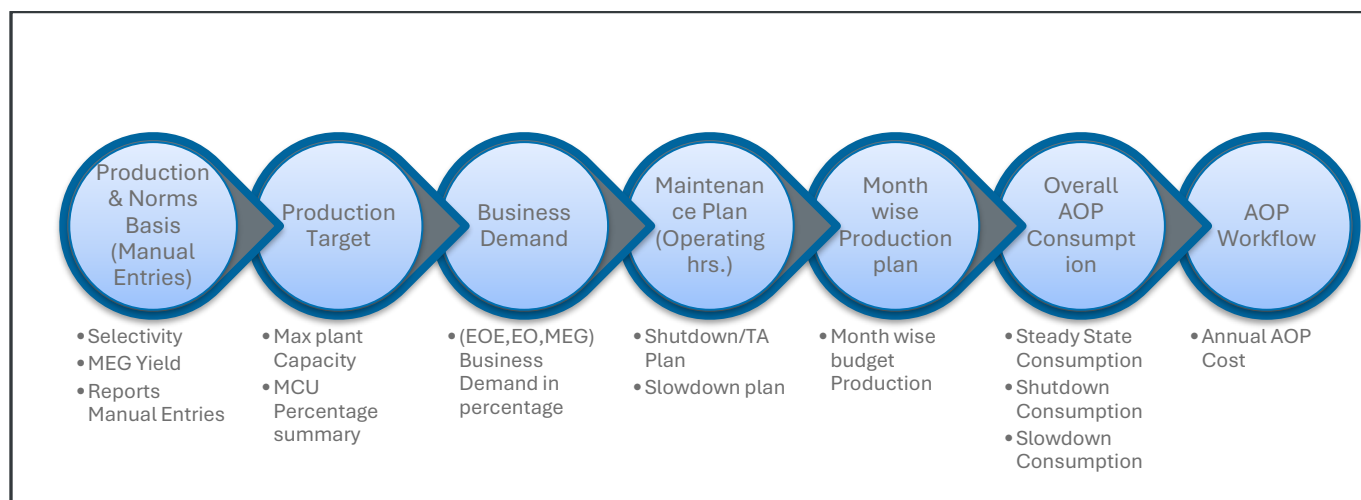


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Month wise Production plan

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Production & Norms Basis

The production & norms basis screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data, including Yield, Selectivity, and Consumption, are configured under the production & norms basis screen to support Production and Overall AOP Consumption calculations and reports generation.

By default, the values entered for the current year will be automatically carried forward to the next year.

The user must insert the TST value month wise and save it with a valid reason. This will provide the Latent heat against TST.

NOTE- Latent heat will be calculated only if the user provides the TST temperature; otherwise, the latent heat value will be set to 0.

In production & norms basis screen, below additional features and tabs are provided.

- AOP historical period basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- Constant tab is provided to enter the constant value which will not change month wise as per below snapshot.
- Report manual entry tab is provided to enter report specific manual entries for last budgeted FY actual values (e.g. Budgeted selectivity, Actual Selectivity, guaranteed selectivity and actual operating Hours)
- Option to import and export manually entered from production & norms basis screen is provided.

The screenshot below highlights the details captured in the digitized AOP platform.

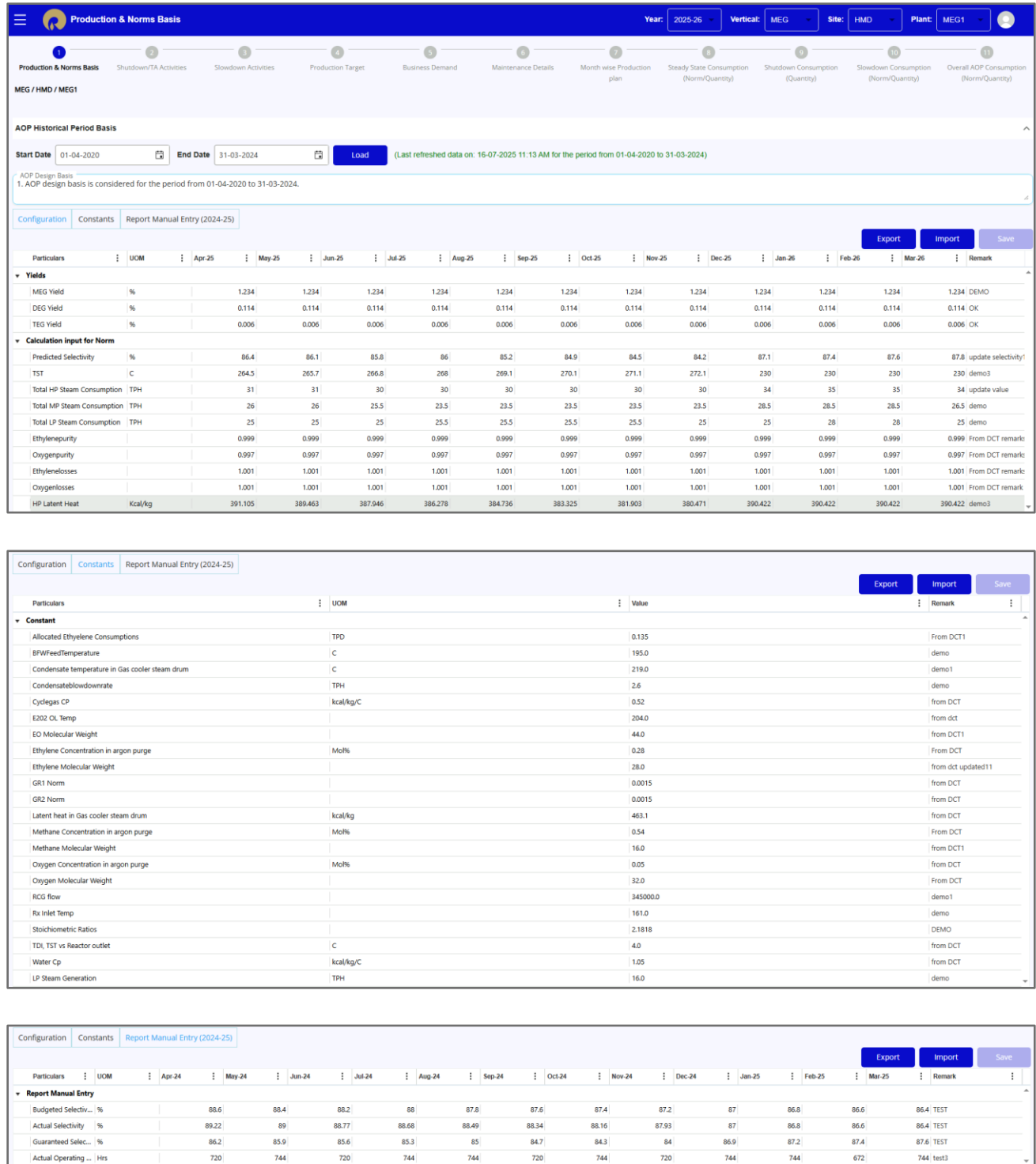


Figure 2 Digitized AOP screen for Production & Norms Basis

2.1.1.2 Production Target

The Max Capacity (MC) of MEG plants is defined on an EOE basis.

MC values for Ethylene Oxide (EO), Ethylene Oxide Equivalent (EOE) and Monoethylene Glycol (MEG) will be captured separately.

Month wise data is considered for deriving the best value. This will account for seasonal effects as well as the festive season.

If AOP basis is considered as 05 years, value estimation involves averaging production data over continuous 5-day periods in the last 5 years, after removing outliers and considering the normal operating mode without recycle for each month, 5 such averages are calculated and the maximum average value from the available 05 average set is designated as the MC value or production volume. Also, the edit option provided to change the derived MC value.

NOTE- AOP historical period basis added in production & norms basis screen where user can change Start date and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

A similar approach is employed for calculating the EO MC value.

The following logic has been formulated to derive month-wise EOE and EO MC values.

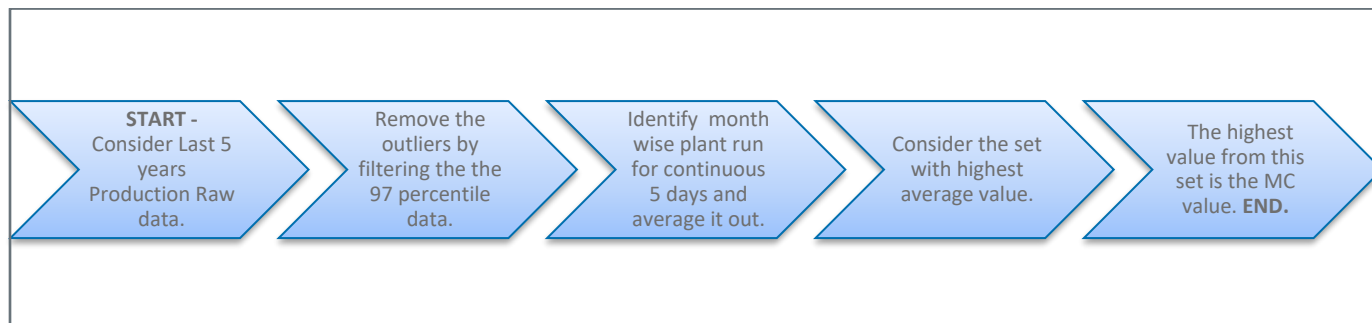


Figure 3 Digitized AOP Steps to derive EOE and EO month wise MC values

The MEG MC value is determined using the following formula.

$$MEG\ MC = MEG\ Yield * (EOE\ MC - EO\ MC)$$

Where MEG yield is a month-wise manual entry captured in the AOP digitization portal in the Production & norms basis screen, as seen in the screenshot below.

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	%		1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234 Yield value
DEG Yield	%		0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114 test
TEG Yield	%		0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006 Testing for UAT

Figure 4 Digitized AOP screen for MEG Yield

If the user needs to modify the production rate this can be done by providing a specific reason.

The AOP digitization portal captures a month-wise percentage summary, comparing each month's value against the highest MU value recorded over the past 12 months.

This information is captured in the AOP digitization portal as seen in the screenshot below.

Production Target

Year: 2025-26 Vertical: MEG Site: HMD Plant: MEG1 2

1

2

3

4

5

6

7

8

9

10

11

Production & Norms Basis

Shutdown/TA Activities

Slowdown Activities

Production Target

Business Demand

Maintenance Details

Month wise Production plan

Steady State Consumption (Norm/Quantity)

Shutdown Consumption (Quantity)

Slowdown Consumption (Norm/Quantity)

Overall AOP Consumption (Norm/Quantity)

MEG / HMD / MEG1

Export

Import

Save

Calculate

Select UOM
TPH

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
EOE	19.167	19.167	18.958	18.958	18.958	18.958	18.958	18.958	18.458	19.583	19.792	19.792	19.583 demo1
EO	4.167	4.167	4.167	4.167	4.167	4.167	4.167	4.037	3.907	4.167	4.167	4.167	updated
MEG	18.513	18.513	18.256	18.256	18.256	18.256	18.256	18.416	17.96	19.027	19.284	19.284	19.027 update

Percentage Summary

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
EOE	96.84	96.84	95.79	95.79	95.79	95.79	95.79	95.79	93.26	98.95	100	100	98.95
EO	100	100	100	100	100	100	100	96.88	93.76	100	100	100	100
MEG	96	96	94.67	94.67	94.67	94.67	94.67	95.5	93.13	98.67	100	100	98.67

Figure 5 Digitized AOP screen for Production Target

The '**Calculate**' button on this screen allows the user to compute the MEG MC value if any changes have been made to the MEG Yield and Start and End date for AOP basis selection on the production & norms basis screen.

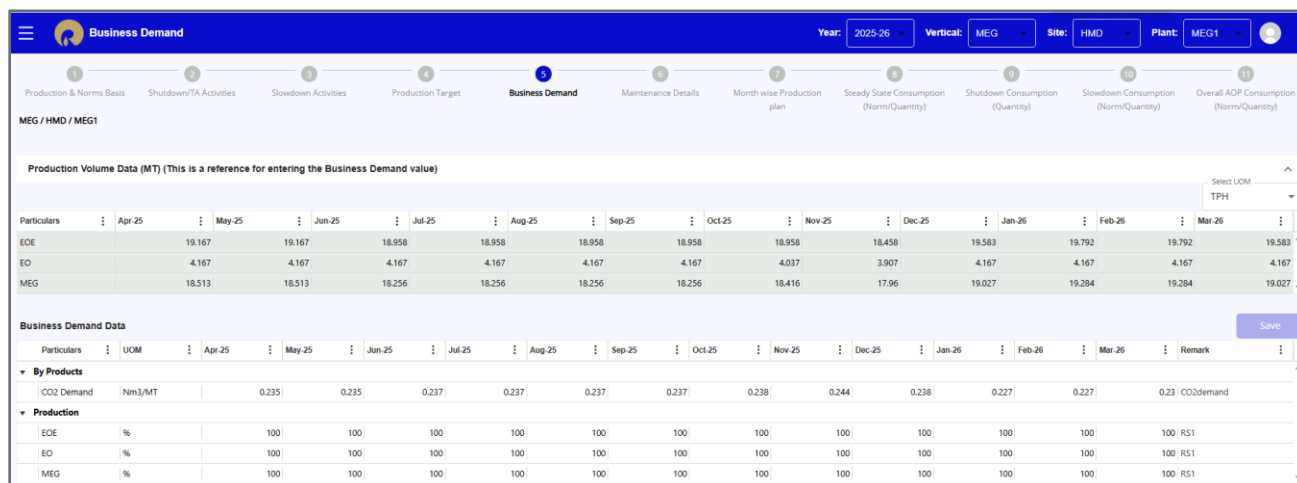
2.1.1.3 Business Demand

The production demand is considered as the maximum value from last year's achieved production values. The default value for EOE, EO and MEG is 100%.

The business demand value can be modified by users upon providing a specific reason.

This information is captured in the AOP digitization portal as seen in the screenshot below.

NOTE- For the HMD MEG plant, CO2 demand will be captured in this section.



Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
EOE	19.167	19.167	18.958	18.958	18.958	18.958	18.958	18.958	18.458	19.583	19.792	19.583
EO	4.167	4.167	4.167	4.167	4.167	4.167	4.167	4.037	3.907	4.167	4.167	4.167
MEG	18.513	18.513	18.256	18.256	18.256	18.256	18.256	18.416	17.96	19.027	19.284	19.027

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
By Products														
CO2 Demand	Nm3/MT	0.235	0.235	0.237	0.237	0.237	0.237	0.237	0.238	0.244	0.238	0.227	0.227	0.23 CO2demand
Production														
EOE	%	100	100	100	100	100	100	100	100	100	100	100	100	RS1
EO	%	100	100	100	100	100	100	100	100	100	100	100	100	RS1
MEG	%	100	100	100	100	100	100	100	100	100	100	100	100	RS1

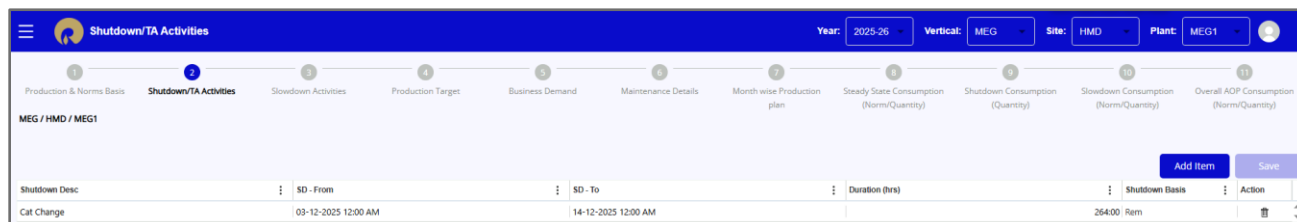
Figure 6 Digitized AOP screen for Business Demand

2.1.1.4 Maintenance/Shutdown/Slowdown/TA Activities

The operations team will provide the shutdown, turnaround and slowdown plan to calculate month-wise operating hours available for production.

The below screenshot depicts the shutdown and turnaround activities planned, as will be captured in the AOP digitized platform.

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration(hrs) to calculate the SD-To date/time.



Shutdown Desc	SD-From	SD-To	Duration (hrs)	Shutdown Basis	Action
Cat Change	03-12-2025 12:00 AM	14-12-2025 12:00 AM		26400 Rem	

Figure 7 Digitized AOP screen for Shutdown/Turnaround Activities

The Slowdown entries will have the inputs as shutdowns, along with the requirement of plant EOE/EO load (TPH basis) for the duration of the slowdown.

The slowdown activities can be calculated based on the slowdown rate and hours. This is called a Basis Shutdown, equivalent to Shutdown.

$$\text{Slowdown rate(TPH)} = (\text{MCU} - \text{Operating Production rate})$$

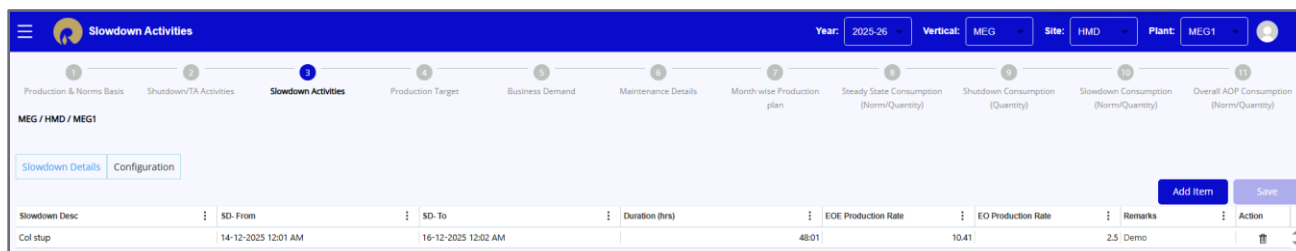
$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate(TPH)} * \text{Slowdown hrs}) / \text{MCU}$$

Operating Production rate will be a manual entry in Slowdown activity screen and slowdown rate will

be calculate at the backend as per above formula.

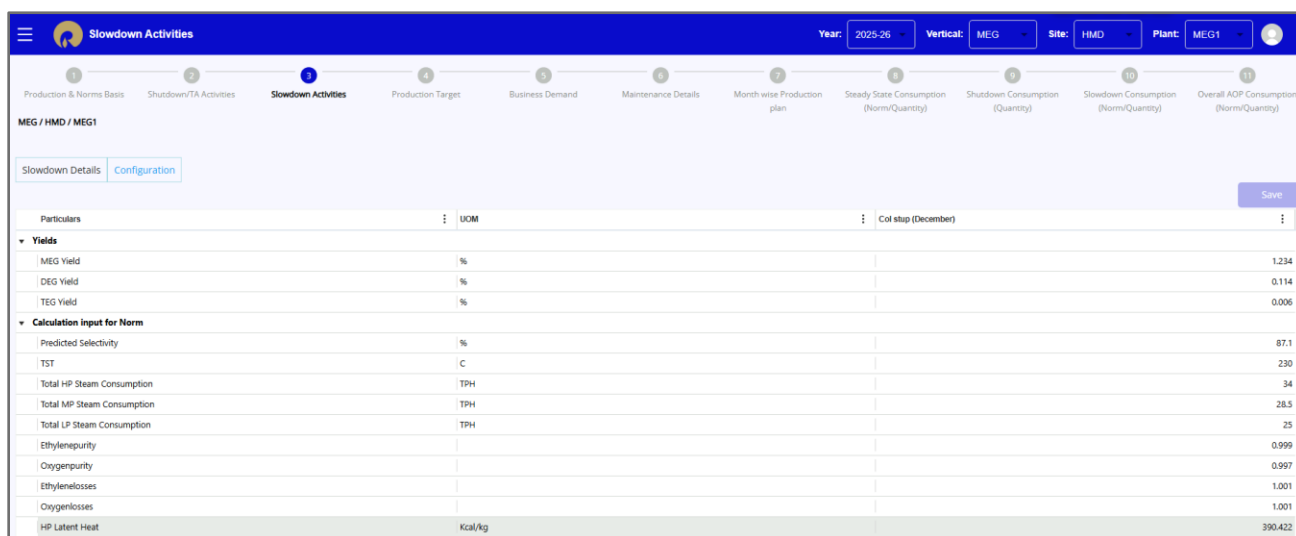
For each shutdown entry, two slowdown entries will be generated automatically, with the nomenclature as “#Shutdown Entry name. – Ramp down” and “#Shutdown Entry name. – Ramp Up”.

In Slowdown details tab, operating production rate-EOE (TPH) and operating production rate-EO (TPH) these two columns will be added in Slowdown activities screen as shown below.



Slowdown Desc	SD-From	SD-To	Duration (hrs)	EOE Production Rate	EO Production Rate	Remarks	Action
Col stop	14-12-2025 12:01 AM	16-12-2025 12:02 AM		48.01	10.41	2.5 Demo	

Separate tab (Configuration) will be provided as shown below in slowdown activity screen for the provision of manual entry related to production & norms basis screen parameters based on the type of slowdown. Number of rows mentioned in slowdown activity screen will be no of column for slow down manual entry screen.



Particulars	UOM	Col stop (December)
Yields		
MEG Yield	%	1.234
DEG Yield	%	0.114
TEG Yield	%	0.006
Calculation input for Norm		
Predicted Selectivity	%	87.1
TST	C	230
Total HP Steam Consumption	TPH	34
Total MP Steam Consumption	TPH	28.5
Total LP Steam Consumption	TPH	25
Ethylene purity		0.999
Oxygen purity		0.997
Ethylene losses		1.001
Oxygen losses		1.001
HP Latent Heat	Kcal/kg	390.422

Figure 8 Digitized AOP screen for Slowdown Activities

The 'Add Item' button on the Shutdown/Slowdown activities screen allows users to insert a new entry for shutdown/slowdown.

The 'Save' button enables users to save newly added items as well as any modifications made to existing entries.

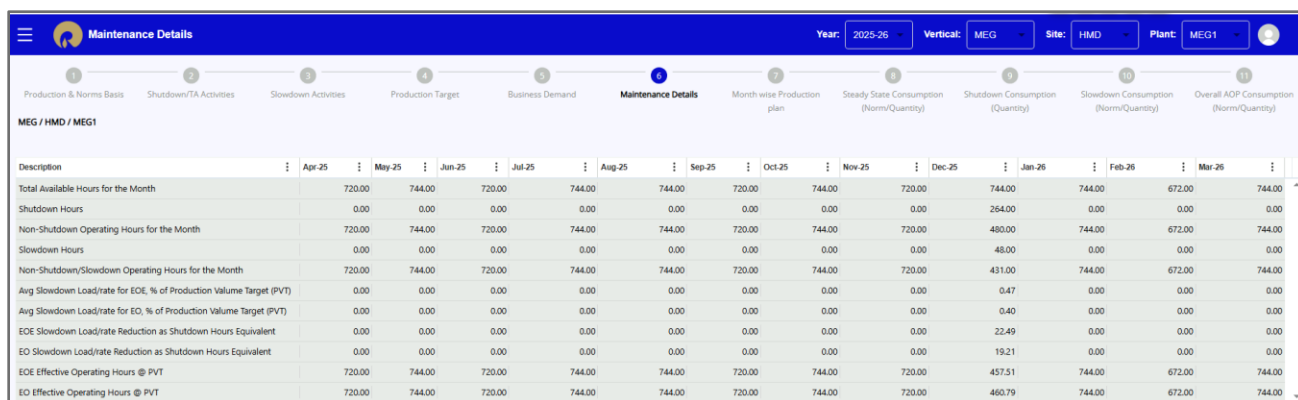
For each month, the following table is generated based on all inputs for shutdowns/slowdowns-

Description	April	...	March
Total Available Hours for the Month	$30 \times 24 = 720$		
Shutdowns Hours	<i>From the Routine Shutdown Plan Section</i>		
Non-Shutdown Operating Hours for the Month	<i>Total Available operating hours for the month – Shutdown hrs</i>		
Slowdown Hours	<i>From the Slowdown activities details</i>		
Non-Shutdown/Slowdown Operating Hours for the Month	<i>Non Shutdown Operating Hours for the Month – Slowdown Hours</i>		
Average Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EOE \text{ slowdown } hrs_i \times EOE \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EOE \text{ slowdown } hrs_i) \times EOE \text{ MC for Month}}$		
Average Slowdown Load/rate for EO, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EO \text{ slowdown } hrs_i \times EO \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EO \text{ slowdown } hrs_i) \times EO \text{ MC for Month}}$		
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	$EOE \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EOE \text{ slowdown } hrs_i)$		
EO Slowdown Load/rate Reduction as Shutdown Hours Equivalent	$EO \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EO \text{ slowdown } hrs_i)$		
EOE Effective Operating Hours @ PVT	<i>Total Available operating hours for the month – Shutdown/TA hours – EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EOE Slowdown load reduction as Shutdown hours Equivalent (Rampup / RampDown)</i>		
EO Effective Operating Hours @ PVT	<i>Total Available operating hours for the month – Shutdown/TA hours – EO Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EO Slowdown load reduction as Shutdown hours Equivalent (Rampup / RampDown)</i>		

Table 3 Maintenance Details

The screen below depicts the Maintenance Details Table, which is captured in the digitized platform.

NOTE- The below screen is read-only. The users cannot change the values.



Description	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Total Available Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00
Shutdown Hours	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	264.00	0.00	0.00	0.00
Non-Shutdown Operating Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	480.00	744.00	672.00	744.00
Slowdown Hours	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.00	0.00	0.00	0.00
Non-Shutdown/Slowdown Operating Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	431.00	744.00	672.00	744.00
Avg Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.47	0.00	0.00	0.00
Avg Slowdown Load/rate for EO, % of Production Volume Target (PVT)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.40	0.00	0.00	0.00
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.49	0.00	0.00	0.00
EO Slowdown Load/rate Reduction as Shutdown Hours Equivalent	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.21	0.00	0.00	0.00
EOE Effective Operating Hours @ PVT	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	457.51	744.00	672.00	744.00
EO Effective Operating Hours @ PVT	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	460.79	744.00	672.00	744.00

Figure 9 AOP Maintenance Details Table

2.1.1.5 Calendar Year Based Production Calculations Details/Month wise Production plan

Using the Business Demand, Production Target and Maintenance Plan (Operating Hours), the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$\text{Production AOP (MT)} = (\text{Operating hours for EOE/EO in a month (hrs)} * \text{Percentage of product demand} * \text{MC Value of product(TPH)}) / 100$$

$$\text{Operating hours for MEG} = \text{Total Available hours of the month} - \text{Shutdown/TA hours of the month}$$

$$\text{Operating hours for EOE} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

$$\text{Operating hours for EO} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

Hence, month-wise production is,

$$\text{Total Budgeted Production} = \sum \text{All the values for the month}$$

The month-wise budget production is further used to derive the annual plan in KT, and this is part of the final report.

This information is captured in the AOP digitization portal as seen in the screenshot below.

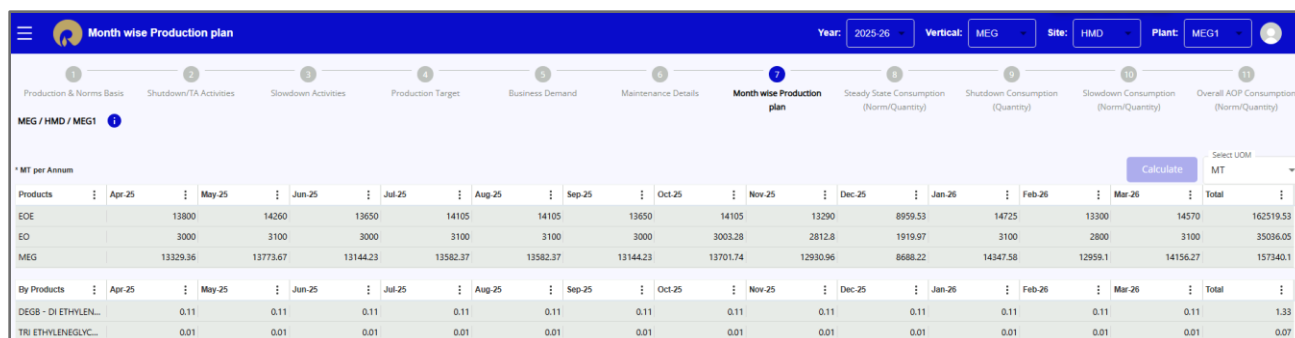


Figure 10 Digitized AOP screen for Month wise Production plan

The **'Calculate'** button on this screen allows users to recalculate Production values when changes are made to production & norms basis settings (e.g., MEG yield), business demand, or when line items are added to slowdown and shutdown activities.

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Steady State Consumption

All the specific Overall AOP Consumption are set with One-unit EOE as the basis.

For calculation purpose in the Digitized AOP platform, based on the DCT provided by the user, Norms are broadly classified as –

1. Historical Best
2. Based on Calculations
3. Manual Entries

3.1.2.1.1 Historical Best

Based on the historical data of last 5 years, corresponding to the MCU production, the best-achieved feed slate period is filtered. This data is further used for the calculation of consumption norms, which is the average of the norms during that period, excluding the identified outliers.

Below is the logic formulated to derive the Historical best Normal Operation Norms –

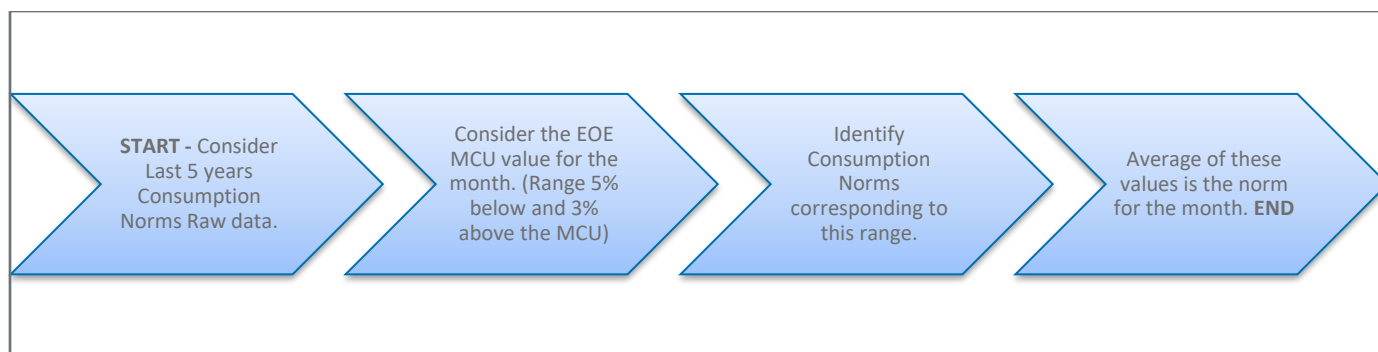


Figure 11 Digitized AOP Steps to derive month wise Historical Best Norms

The user has provision to overwrite/update the value by providing a valid reason for the historical best particulars.

The list of particulars to be derived using the Historical Best logic is specified in the DCT provided by the user. These particulars may vary from plant to plant.

3.1.2.1.2 Based on Calculations

For the MEG vertical, particulars like ethylene, oxygen, HP Steam, BFW water etc. are formulated based on the stoichiometric calculations provided by the user in the DCT.

The formula can vary from plant to plant.

E.g. the Raw Material norms for ethylene and oxygen for HMD MEG are derived using the calculations mentioned in the below table.

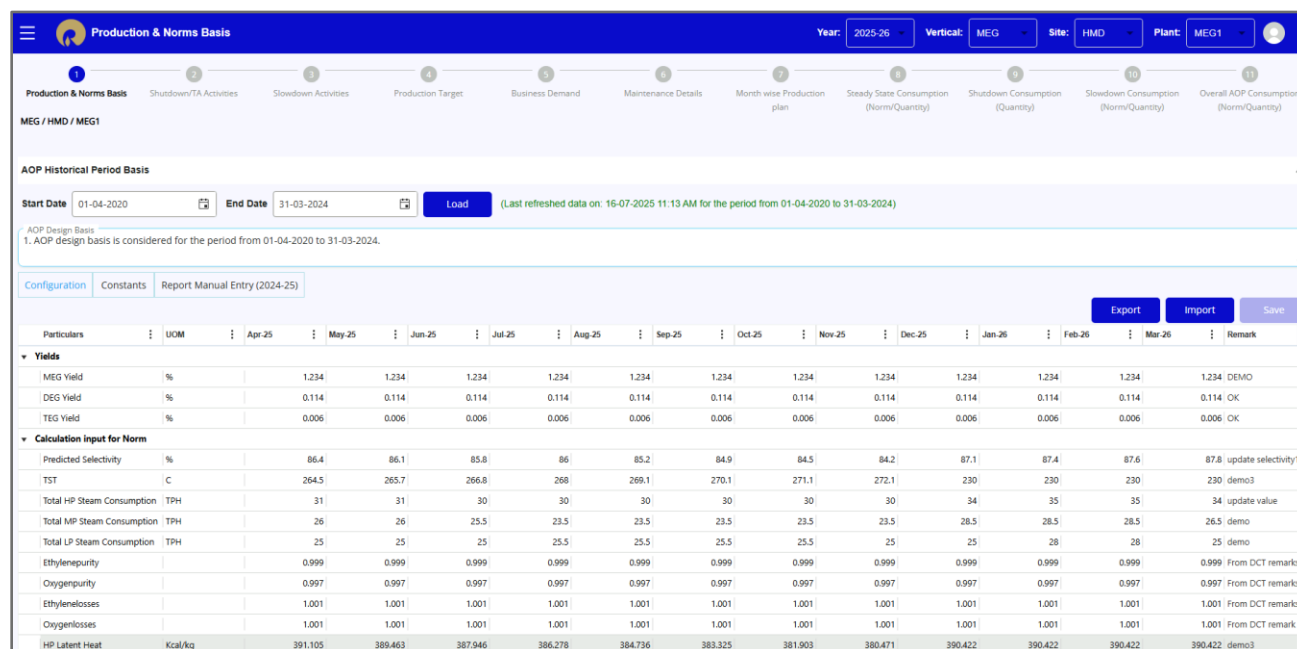
Plant	Category	Particular	Calculation
HMD MEG	Raw Material	Ethylene	$\left(\frac{1.001 \times 28 \times 100}{44 \times \{selectivity\}} + \frac{[METHANE] \times 28 \times 0.28}{16 \times 0.54} \right) / 0.999 + (0.135 / 24 \times ([EOE] / [EOE \text{ effective Operating hrs}]))$ Where, 1.001 - decimals = EOE Losses 28 - Ethylene Molecular Weight 44 - EO molecular weight 0.28 - Ethylene concentration in argon purge (Mol %) 16 - Methane molecular weight 0.54 - Methane concentration in argon purge (Mol %) 0.999 - Ethylene Purity (wt%) 0.135 - 1.9 - Allocated Ethylene consumptions (TPD)

Plant	Category	Particular	Calculation
		Oxygen	$(1.001 * (100 * 2.1818 / \{selectivity\} - 1.818) + ([METHANE] * 32 * 0.05 / 16 / 0.54)) / 0.997$ Where, 1.001 - decimals = EOE Losses 2.1818, 1.881 - Stoichiometric Ratios 32 - Oxygen molecular weight 0.05 - Oxygen concentration in argon purge (Mol %) 16 - Methane molecular weight 0.54 - Methane concentration in argon purge (Mol %) 0.997 - Oxygen Purity (wt%)

Table 4 Raw Material Norms calculations

The values for *losses and purity* can be captured from the Lab analysis or Past/last year's operations. The user has the provision to enter the value manually in the Production & norms basis screen.

Selectivity, Losses and Purity can be modified/added by the user in the Production & norms basis Screen under the Calculation Input for Norms Subsection.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	%	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234	1.234 DEMO
DEG Yield	%	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114	0.114 OK
TEG Yield	%	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006 OK
Calculation input for Norm														
Predicted Selectivity	%	86.4	86.1	85.8	86	85.2	84.9	84.5	84.2	87.1	87.4	87.6	87.8	update selectivity
TST	C	264.5	265.7	266.8	268	269.1	270.1	271.1	272.1	230	230	230	230	demo3
Total HP Steam Consumption	TPH	31	31	30	30	30	30	30	30	34	35	35	34	update value
Total MP Steam Consumption	TPH	26	26	25.5	23.5	23.5	23.5	23.5	23.5	28.5	28.5	28.5	26.5	demo
Total LP Steam Consumption	TPH	25	25	25	25.5	25.5	25.5	25.5	25	25	28	28	25	demo
Ethylenepurity		0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	From DCT remark
Oxygenpurity		0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	From DCT remark
Ethylenelosses		1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	From DCT remark
Oxygenlosses		1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	1.001	From DCT remark
HP Latent Heat	Kcal/kg	391.105	389.463	387.946	386.278	384.736	383.325	381.903	380.471	390.422	390.422	390.422	390.422	demo3

Figure 12 Digitized AOP screen for Catalyst Selectivity month wise

In the Steady State Consumption screen the cells for particulars with formula, e.g. HP Steam Dis, will be read only. If the values for such particulars are to be modified then the respective inputs should be modified, e.g. Total HP Steam consumption in the Production & norms basis Screen.

3.1.2.1.3 Manual Entries

The particulars under the Chemicals and Catalyst sections are manually entered by the user, who must also provide the corresponding norm values in the Normal Operation Norms screen.

The screenshot below highlights the details captured in the digitized AOP platform in the Steady State Consumption screen.

Steady State Consumption (Norm/Quantity)

Year: 2025-26 Vertical: MEG Site: HMD Plant: MEG1

1

2

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Production & Norms Basis

Shutdown/T/A Activities

Slowdown Activities

Production Target

Business Demand

Maintenance Details

Month wise Production plan

Steady State Consumption (Norm/Quantity)

Shutdown Consumption (Quantity)

Slowdown Consumption (Norm/Quantity)

Overall AOP Consumption (Norm/Quantity)

MEG / HMD / MEG1

Export

Import

Save

Calculate

Particulars	UOM / MT	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26
▼ Raw Material												
Ethylene	MT		0.742	0.744	0.747	0.745	0.752	0.755	0.758	0.761	0.736	0.733
OXYGEN - NP01	MT		0.711	0.72	0.728	0.723	0.746	0.756	0.768	0.777	0.69	0.682
▼ By Products												
CO2			0	0	0	0	0	0	0	0	0	0
DEGB - DI ETHYLENE GLYCOL	MT		0.11	0.11	0.11	0.11	0.11	0.11	0.111	0.111	0.111	0.111
GLYCOL RESIDUE I	MT		0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002
GLYCOL RESIDUE II	MT		0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002
Process Vent Gas	MT		0.009	0.009	0.009	0.009	0.009	0.009	0.009	0.009	0.01	0.007
TRI ETHYLENEGLYCOL B	MT		0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006
▼ Cat Chem												
CHEM CATAcarb-922	KG		0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007
CHEM PLURONIC PE 10100	KG		0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012
CHEM TRISODIUM PHOSPHATE	KG		0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003
CSUVE - CAUSTIC SODA LYE - GRADE 1	MT		0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
EDC - Ethylene Di Chloride	MT		0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027
POTASSIUM CARBONATE TECH.GRD-02% CL	MT		0	0	0	0	0	0	0	0	0	0
▼ Utility Consumption												
Boiler Feed Water	m3		1.578	1.618	1.636	1.571	1.719	1.764	1.832	1.817	2.339	2.288
Cooling Water_Dis	Km3		0.417	0.417	0.422	0.422	0.422	0.422	0.423	0.434	0.487	0.404
D M WATER	m3		0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.1	0.012
Effluent Treated	m3		1.696	1.696	1.714	1.714	1.714	1.714	1.761	1.761	2.246	1.642
HP Nitrogen	Nm3		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
HP Steam_Dis	MT		0.123	0.083	0.031	0.096	0	0	0	0	0	0
Instrument air	Nm3		19.461	19.461	19.675	19.675	19.675	19.675	19.701	20.212	23.073	18.846
LP Nitrogen	Nm3		32.609	32.609	32.967	32.967	32.967	32.967	33.011	33.868	40.839	31.579
LP STEAM_DIS	MT		0.47	0.47	0.435	0.351	0.276	0.211	0.123	0.143	0.46	0.606

Figure 13 Digitized AOP screen for Steady State Consumption

The Normal Operation Normas screen also captures the Intermediate calculations required for Norms calculation as shown in the below screen shot –

Intermediate Values														
Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
▼ Calculated Intermediate Values														
CO2 Vent	Nm3	4340.26443	4599.85815	4513.84204	4587.91848	4895.64185	4850.83097	5169.67853	4982.92384	2497.6381	4241.56987	3761.65069	4045.15927	
CO2 Generation	Nm3	4340.49943	4600.09315	4514.07904	4588.15548	4895.87885	4851.06797	5169.91653	4983.16684	2497.8761	4241.79687	3761.87769	4045.38927	
Excess.HP	TPH	0	0	0	0	0.99212	1.85001	3.13105	1.93657	10.20789	8.69303	7.82381	7.19417	
Excess.MP	TPH	0	0	0.37924	2.84411	4.26236	5.50763	7.17607	6.36897	0	0	0	0	
Heat.to.HP.Steam	kcal/h	13295106.762	13634944.822	13630357.818	13048033.25	14335267.145	14720555.108	15300309.578	14736487.146	18884441.602	18664505.904	18293200.534	17597062.688	
Heat.to.MP.Steam	TPH	11571300	11786580	11983920	12199200	12396540	12575940	12755340	12934740	5382000	5382000	5382000	5382000	
HP.Steam.Generation	TPH	28.64808	29.40454	29.41761	28.1813	30.99212	31.85001	33.13105	31.93657	44.20789	43.69303	42.82381	41.19417	
MP.Steam.Generation	TPH	24.9883	25.45317	25.87924	26.34411	26.77024	27.15762	27.54501	27.9324	11.62353	11.62358	11.62358	11.62353	
Reaction.Heat	kcal/h	32580606.762	33135724.822	33328477.818	32959433.25	34446007.145	35010695.108	35769849.578	35385427.146	31980641.602	31760705.904	31389400.534	30693262.688	

Figure 14 Digitized AOP screen for Steady State Consumption (Calculated Intermediated values)

The **'Calculate'** button on the screen will enable if the user makes any change in the dependent inputs in the Production & norms basis screen.

The **'Save'** button on the above screen enables the user to save any norm value updated manually for historical best norms for cat chem.

2.1.2.2 Shutdown Consumption

The Shutdown Consumption are recorded for the months in which shutdown activities are planned,

as specified in the Shutdown Activities screen.

The shutdown quantity utilization data includes TA, asset maintenance linked shutdown/ slowdown activities, etc..

The user must enter the consumption data (absolute values) for each line item.

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.

Shutdown Consumption (Quantity)

Year:2025-26

Vertical:MEG

Site:HMD

Plant:MEG1

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Production & Norms Basis

Shutdown/TA Activities

Slowdown Activities

Production Target

Business Demand

Maintenance Details

Month wise Production plan

Steady State Consumption (Norms/Quantity)

Shutdown Consumption (Quantity)

Slowdown Consumption (Norms/Quantity)

Overall AOP Consumption (Norms/Quantity)

MEG / HMD / MEG1

Save

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
▼ Raw Material														
Ethylene	MT													Click to add remark
OXYGEN - NP01	MT													Click to add remark
▼ By Products														
CO2														Click to add remark
DEGB - DI ETHYLENE GLYCOL	MT													Click to add remark
GLYCOL RESIDUE I	MT													Click to add remark
GLYCOL RESIDUE II	MT													Click to add remark
Process Vent Gas	MT													Click to add remark
TRI ETHYLENEGLYCOL B	MT													Click to add remark
▼ Cat Chem														
CHEM CATACARB-922	KG	0	0	0	0	0	0	0	0	0	0	0	0	Click to add remark
CHEM PLURONIC PE 10100	KG													Click to add remark
CHEM TRISODIUM PHOSPHATE	KG													Click to add remark
CSLYE - CAUSTIC SODA LYE - GRADE 1	MT													Click to add remark
EDC - Ethylene Di Chloride	MT													Click to add remark
POTASSIUM CARBONATE TECH.GRD-.02% CL	MT													Click to add remark
▼ Utility Consumption														
Boiler Feed Water	m3	0	0	0	0	0	0	0	0	0	0.52	0	0	0 ok
Cooling Water_Dis	Km3	0	0	0	0	0	0	0	0	0	2400	0	0	0 update12
D M WATER	m3	0	0	0	0	0	0	0	0	0	80	0	0	0 Test1
Effluent Treated	m3	0	0	0	0	0	0	0	0	0	19.97	0	0	0 test
HP Nitrogen	Nm3													Click to add remark
HP Steam_Dis	MT	0	0	0	0	0	0	0	0	0	0.79	0	0	0 ok
Instrument air	Nm3	0	0	0	0	0	0	0	0	0	12	0	0	0 Test

Figure 14 Digitized AOP screen for Shutdown Consumption

2.1.2.3 Slowdown Consumption

Slowdown Consumption are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The values for Slowdown Consumption are same as the Steady State Consumption for the month.

NOTE- Slowdown wise norms calculation will be provided by RIL.

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.

Slowdown Consumption (Norm/Quantity)

Year: 2025-26Vertical: MEGSite: HMDPlant: MEG1

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Production & Norms Basis

Shutdown/TA Activities

Slowdown Activities

Production Target

Business Demand

Maintenance Details

Month wise Production plan

Steady State Consumption (Norm/Quantity)

Shutdown Consumption (Quantity)

Slowdown Consumption (Norm/Quantity)

Overall AOP Consumption (Norm/Quantity)

MEG / HMD / MEG1

Save

Calculate

Particulars	UOM	Col stop (December)
▼ Raw Material		
Ethylene	MT	0.736
OXYGEN - NP01	MT	0.69
▼ By Products		
CO2		0
DEGB - DI ETHYLENE GLYCOL	MT	0.087
GLYCOL RESIDUE I	MT	0.002
GLYCOL RESIDUE II	MT	0.002
Process Vent Gas	MT	0.01
TRI ETHYLENEGLYCOL B	MT	0.005
▼ Cat Chem		
CHEM CATACARB-922	KG	
CHEM PLURONIC PE 10100	KG	
CHEM TRISODIUM PHOSPHATE	KG	
CSLYE - CAUSTIC SODA LVE - GRADE 1	MT	
EDC - Ethylene Di Chloride	MT	
POTASSIUM CARBONATE TECH.GRD-.02% CL	MT	
▼ Utility Consumption		
Boiler Feed Water	m3	1.233
Cooling Water_Dis	Km3	0.917
D M WATER	m3	0.189
Effluent Treated	m3	4.225
HP Nitrogen	Nm3	0.019
HP Steam_Dis	MT	2.186
Instrument air	Nm3	43.405

Figure 15 Digitized AOP screen for Slowdown Consumption

2.1.2.4 Budget Norms/Overall AOP Consumption

Using the values of Steady State Consumption, Shutdown Consumptions and Slowdown Consumption, the final Overall AOP Consumption value is calculated as below –

Consumption AOP for months without shutdown or slowdown = Normal Operations Norms

*Consumption AOP for months with shutdown or slowdown = Total Consumption/
Total production*

Where,

*Total Consumption = (Normal Operation Norms * EOE MC value * EOE operating hours) +
 $\sum_{i=1}^n (\text{Slowdown Norms} * \text{EOE slowdown hours} * \text{EOE slowdown rate})_n + \text{Shutdown Consumption}$*

*Total Production = (EOE MC value * EOE operating hours) + $\sum_{i=1}^n (\text{EOE slowdown hours} * \text{EOE slowdown rate})_n$*

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.

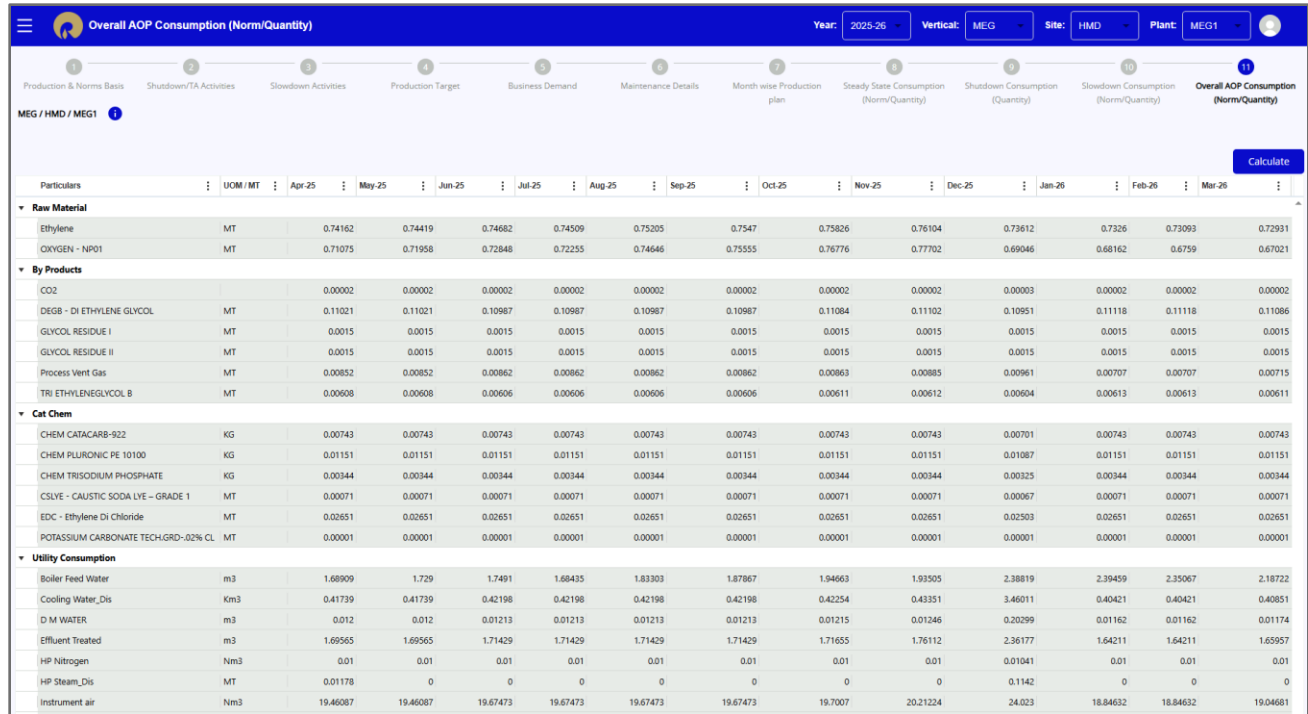


Figure 16 Digitized AOP screen for Budget Norms or Overall AOP Consumption

3. AOP Reports

- Dataset (Basis)
- Annual AOP Cost.
- Plant Production Summary (T-14).
- Month wise Production Plan (T – 16)
- Month wise Raw data (T – 18).
- Turnaround Report (T – 19A).
- Annual Production Plan (T – 15).
- Plant Contribution (T – 21).

The below screenshot depicts the AOP report captured in the AOP digitized platform.

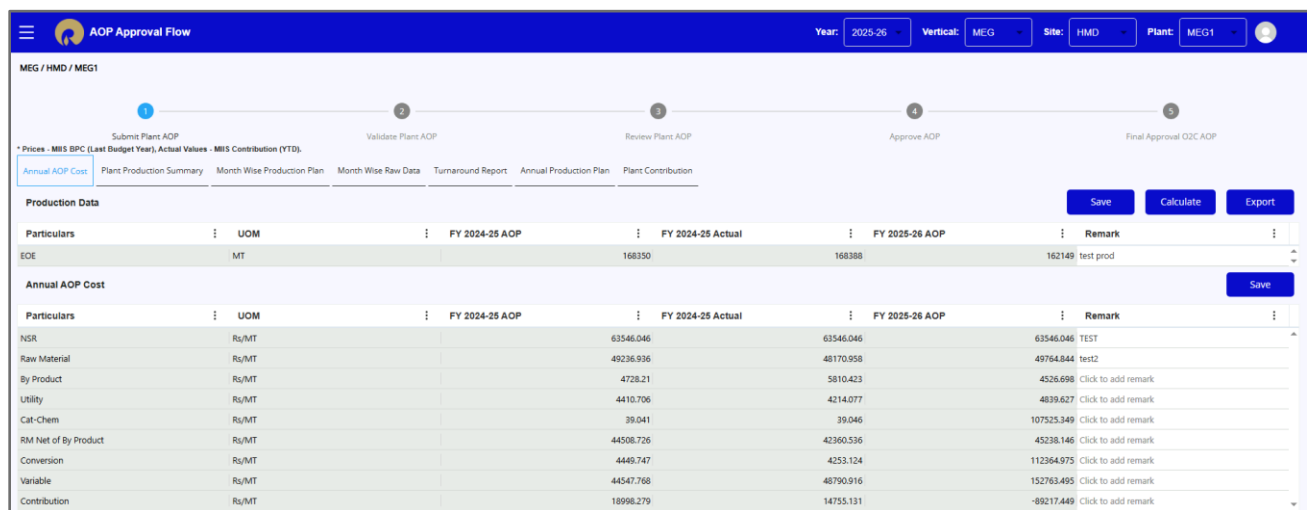


Figure 17 Digitized AOP screen for Annual AOP Cost

AOP Approval Flow										
MEG / HMD / MEG1										
1 2 3 4 5 Submit Plant AOP Validate Plant AOP Review Plant AOP Approve AOP Final Approval Q2C AOP										
Annual AOP Cost Plant Production Summary Month Wise Production Plan Month Wise Raw Data Turnaround Report Annual Production Plan Plant Contribution										
Plant Production Summary (T-14)										
SL.No	Item	Unit	2024-25		2025-26	Variance wrt current year budget		Variance wrt current year actuals		Remarks
	Production Volume		Budget	Actual	Budget	MT	%	MT	%	
1	EOE	MT	168350.184	168388.233	162148.865	-6201.319	-3.684	-6239.368	-3.705	Click to add remark
2	MEG	MT	163080	159509.078	156748.91	-6331.09	-3.882	-2780.168	-1.73	Click to add remark
3	EO	MT	36216	36165.786	35144.4	-1071.6	-2.959	-1021.386	-2.824	Click to add remark
4	Production	MT	168350.184	168388.233	162148.865	-6201.319	-3.684	-6239.368	-3.705	Click to add remark
5	Operating Hours	Hrs	8760	8760	8496	-264	-3.014	-264	-3.014	Click to add remark
6	ProductionRate	TPH	19.218	19.222	19.439	0.221	1.148	0.216	1.125	Click to add remark
7	NSR	Rs/MT	63546.046	63546.046	63546.046	0	0	0	0	Click to add remark
8	Raw Material	Rs/MT	49236.936	48170.958	49764.844	-527.908	-1.072	-1593.886	-3.309	Click to add remark
9	By Product	Rs/MT	4728.21	5810.423	4526.698	-201.512	-4.262	-1283.725	-22.093	Click to add remark
10	RM Net of By Product	Rs/MT	44508.726	42360.536	45238.146	-729.42	-1.639	-2877.611	-6.793	Click to add remark
11	Cat-Chem	Rs/MT	39.041	39.046	107525.349	-107486.307	-27513.815	-107486.302	-275278.886	Click to add remark
12	Utility	Rs/MT	4410.706	4214.077	4839.627	-438.921	-9.725	-625.55	-14.844	Click to add remark
13	Conversion	Rs/MT	4449.747	4253.124	112364.975	-107915.228	-2425.199	-108111.852	-2541.94	Click to add remark
14	Variable	Rs/MT	44547.768	48790.916	152763.495	-108215.727	-242.921	-103972.579	-213.098	Click to add remark
15	Contribution	Rs/MT	18998.279	14755.131	-89217.449	-108215.727	-569.608	-103972.579	-704.654	Click to add remark
16	Contribution Rs Lacs	Rs lacs	31983.637	24845.904	-144665.08	-176648.717	-552.31	-169510.984	-682.249	Click to add remark

Figure 18 Digitized AOP screen for Production Plant Summary

AOP Approval Flow

Year:2025-26Vertical:MEGSite:HMDPlant:MEG1

MEG / HMD / MEG1

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval Q2C AOP

* Prices - M1S BPC (Last Budget Year), Actual Values - M1S Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Production (T-16)

Save

SL...	Month	2024-25						2025-26						Remarks
		EOE Production, MT		Operating Hours		Throughput, TPH		Operating ...	MEG Throu...	EO Throu...	EOE Throu...	Total EOE, ...		
		Budget	Actual	Budget	Actual	Budget	Actual							
1	April	13800	13793	720	720	19.17	19.16	720	18.38	4.48	19.37	13949.59	Click to add remark	
2	May	14260	14284	744	744	19.17	19.2	744	18.34	4.43	19.29	14353.98	Click to add remark	
3	June	13800	13787	720	720	19.17	19.15	720	18.12	4.53	19.22	13836.23	Click to add remark	
4	July	14260	14262	744	744	19.17	19.17	744	18.24	4.41	19.2	14282.1	Click to add remark	
5	August	14260	14320	744	744	19.17	19.25	744	18.42	4.37	19.29	14352.29	Click to add remark	
6	September	13800	13871	720	720	19.17	19.27	720	18.46	4.4	19.35	13934.81	Click to add remark	
7	October	14260	14383	744	744	19.17	19.33	744	18.42	4.45	19.37	14411.6	Click to add remark	
8	November	13800	13917	720	720	19.17	19.33	720	18.45	4.46	19.41	13976.05	Click to add remark	
9	December	14415	14409	744	744	19.38	19.37	440	18.55	4.65	19.67	8656.29	Click to add remark	
10	January	14415	14296	744	744	19.38	19.22	744	18.65	4.61	19.72	14673.62	Click to add remark	
11	February	13020	12953	672	672	19.38	19.28	672	18.71	4.52	19.68	13325.59	Click to add remark	
12	March	14260	14115	744	744	19.17	18.97	744	18.78	4.46	19.68	14641.68	Click to add remark	
13	Total	168350	168390	8760	8760	19.22	19.22	8456	18.46	4.47	19.43	164293.83	Click to add remark	

Main Products - Production for the budget year

Products	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
EOE	13800	14260	13650	14105	14105	13650	14074	13290	8619.86	14725	13300	14570	162148.86
EO	3002.4	3102.48	3002.4	3102.48	3102.48	3002.4	3005.76	2815.2	2001.6	3102.48	2802.24	3102.48	35144.4
MEG	13326.4	13770.61	13141.27	13579.31	13579.31	13141.27	13660.42	12928	8168.26	14344.51	12956.34	14153.21	156748.91

Figure 19 Digitized AOP screen for Month Wise Production Plan

AOP Approval Flow

MEG / HMD / MEG1

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval Q2C AOP

* Prices - MBS BPC (Last Budget Year), Actual Values - MBS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Consumption (T-18)

Parameters	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
Budgeted Selectivity L...	88.6	88.4	88.2	88	87.8	87.6	87.4	87.2	87	86.8	86.6	86.4	
Actual Selectivity for L...	89.22	89	88.77	88.68	88.49	88.34	88.16	87.93	87	86.8	86.6	86.4	
Guaranteed Selectivity...	86.2	85.9	85.6	85.3	85	84.7	84.3	84	86.9	87.2	87.4	87.6	
Predicted Selectivity L...	86.4	86.1	85.8	86	85.2	84.9	84.5	84.2	87.1	87.4	87.6	87.8	

RawMaterial

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
Ethylene	MT	0.740151	10234.315	10612.135	10194.066	10509.48	10607.679	10301.6	10671.695	10113.012	5635.624	10787.55	9721.436	10626.032	120014.624
OXYGEN - NP01	MT	0.716973	9808.309	10261.225	9943.807	10191.525	10528.875	10313.244	10805.511	10326.33	5286.172	10036.899	8989.483	9764.974	116256.353

ByProducts

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
Process Vent Gas	MT	0.0081975	117.594	121.514	117.59	121.51	121.51	117.59	121.407	114.52	73.546	104.175	94.093	104.166	1329.216
TRI ETHYLENEGL...	MT	0.00607745	83.69	86.617	82.659	85.414	85.414	82.659	85.924	81.317	51.013	90.227	81.495	89.024	985.451
DI ETHYLENE GLY...	MT	0.110259	1520.539	1571.224	1499.425	1549.392	1549.392	1499.425	1558.653	1475.084	925.362	1636.713	1478.308	1614.881	17878.398
Carbon Dioxide		1.72397e-005	0.235	0.235	0.237	0.237	0.237	0.237	0.238	0.244	0.211	0.227	0.227	0.23	2.795
GLYCOL RESIDUE I	MT	0.00149108	20.7	21.39	20.475	21.158	21.158	20.475	21.111	19.935	11.484	22.088	19.95	21.855	241.778
GLYCOL RESIDUE II	MT	0.00149108	20.7	21.39	20.475	21.158	21.158	20.475	21.111	19.935	11.484	22.088	19.95	21.855	241.778
CO2			3246.536	3327.643	3267.734	3339.298	3353.444	3255.152	3404.401	3372.174	3498.663	3279.473	2957.411	3325.302	39627.232
CO2 Demand	Nm3/MT		3246.536	3327.643	3267.734	3339.298	3353.444	3255.152	3404.401	3372.174	3498.663	3279.473	2957.411	3325.302	39627.232

CatChem

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
CAUSTIC SODA L...	MT	3.53792	48962.4	50594.48	48962.55	50594.635	50594.635	48962.55	50553.808	48973.65	26581.759	50595.1	46177.6	52116.89	573670.057

Figure 20 Digitized AOP screen for Month Wise Raw Data

AOP Approval Flow

Year:2025-26Vertical:MEGSite:HMDPlant:MEG1

MEG / HMD / MEG1

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval Q2C AOP

* Prices - MBS BPC (Last Budget Year), Actual Values - MBS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Turnaround Details (T-19A)

SL.No	Activities	From	To	Duration, hrs	Remark
1	Cat Change	03-Dec-25	14-Dec-25	26400	Click to add remark

Turnaround details for the previous years data since commissioning

SL.No	Activities	Turnaround Period From	Turnaround Period To	Duration, hrs	Period in Months	Remark	Action
1	Catalyst replacement carried out along with tur...	29-Oct-98	12-Nov-98	351		Click to add remark	
2	Catalyst replacement carried out along with tur...	01-Apr-01	13-Apr-01	253		Click to add remark	
3	EO reactor tube leakage lead to early catalyst c...	29-May-02	13-Jun-02	357		Click to add remark	
4	EO reactor hook up and Capex implementation	24-Aug-04	02-Sep-04	227		Click to add remark	
5	Catalyst replacement carried out along with tur...	22-May-06	30-May-06	238		Click to add remark	
6	Catalyst replacement carried out along with tur...	22-Apr-08	30-Apr-08	216		Click to add remark	
7	Catalyst replacement carried out along with tur...	05-Apr-10	13-Apr-10	216		Click to add remark	
8	Catalyst replacement carried out along with tur...	28-Nov-11	07-Dec-11	218		Click to add remark	
9	Catalyst replacement carried out along with tur...	21-May-13	31-May-13	244		Click to add remark	
10	Catalyst replacement carried out along with tur...	21-Sep-15	29-Sep-15	216		Click to add remark	
11	Catalyst replacement carried out along with tur...	30-Oct-17	08-Nov-17	216		Click to add remark	
12	Catalyst replacement carried out along with tur...	11-Nov-19	25-Nov-19	346		Click to add remark	
13	Catalyst replacement carried out along with tur...	05-Dec-21	16-Dec-21	257		Click to add remark	
14	Catalyst replacement carried out along with tur...	10-Dec-23	20-Dec-23	264		Click to add remark	
15	Cat Change	03-Dec-25	14-Dec-25	264		Click to add remark	

Figure 21 Digitized AOP screen for Turnaround Report

Figure 22 Digitized AOP screen for Annual Production Plan

Figure 23 Digitized AOP screen for Plant Contribution

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by HMD Site team